

Kartik Soni

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SUMMARY

I possess a strong interest in deep learning, low-level systems, and model inference. I am particularly drawn to understanding the internal workings of large language models (LLMs), performance optimization, inference engineering, agentic behaviours, and the infrastructure necessary for efficient model operation. Additionally, I enjoy exploring reinforcement learning (RL) optimization, constructing custom pipelines, and experimenting with open-source models and compression techniques.

EDUCATION

SVKM'S NMIMS MPSTME, Shirpur

Jul 2023 - Present

B.Tech, Artificial Intelligence and Machine Learning

Central Academy Sr. Sec. School — Class 12th

Apr 2022 - Apr 2023

Evergreen Public School — Class 10th

Apr 2020 - Apr 2021

MAJOR PROJECTS

PM2.5 Forecasting (PIML) [Link](#)

Feb 2026 - Apr 2026

- Engineered a Physics Based SpatioTemporal ConvLSTM Model for forecasting PM2.5 concentration fields over India with an accuracy of 89.25%.
- Applied Physics informed WindWarp module to simulate pollutant transport using differentiable grid sampling over 16 hours.

Offline MultiModal RAG [Link](#)

Aug 2025 - Oct 2025

- Built a fully offline RAG System performing semantic search across text, images and audio by eliminating cloud API dependencies for privacy focused deployment.
- Integrated FAISS vector retrieval, Whisper transcription and CLIP-based image embeddings for cross-modal querying.
- Tested over 2000 documents and 500 audio clips, with an accuracy of 73.87%.

FloatChat (Natural Language Query System) [Link](#)

Dec 2025 - Feb 2026

- Developed an intelligent NL query interface for Argo oceanographic datasets, enabling non-technical researchers to explore complex climate data without manual scripting.
- Reduced manual report generation time by 4x, significantly improving researcher workflow efficiency.
- Visualized float trajectories and profile locations with temperature color-coding, detailed T/S (temperature-salinity) diagrams, BGC parameters, and depth profiles.

GPT-2 From Scratch [Link](#)

May 2026 - June 2026

- Implemented the GPT-2 decoder-only transformer architecture from scratch, including multi-head self-attention, positional embeddings, and the training loop.
- Self-directed deep dive into transformer internals to strengthen low-level understanding of LLM architecture beyond using pre-built libraries.

Koala — Vector Search Library [Link](#)

Aug 2026 - Present

- Built a vector search library from scratch, implementing core indexing and nearest-neighbour search algorithms for similarity search.
- Focused on efficient data structures for high-dimensional search, reinforcing core DSA and algorithm-design fundamentals.

SKILLS

- Tools and Visualization:** Power BI, Git, Tableau
- Programming and Databases:** Python, C++, MySQL, SQL Server
- Machine Learning:** PyTorch, TensorFlow, Scikit-Learn, AutoML
- Data Analysis:** Pandas, NumPy, Scipy, NetCDF, XArray
- NLP / LLM:** LangChain, OpenAI API, Claude API, RAG pipelines, Agentic AI, MCP
- Operating Systems:** Working familiarity with Unix/Linux command-line environments
- Soft Skills:** Leadership, Teamwork, Communication, Time Management

CERTIFICATIONS

- Machine Learning with Python Programming — NPTEL
- Deep Learning — IIT Ropar — NPTEL
- Data Analytics — Google
- AWS Builder Essentials — AWS

ACHIEVEMENTS & EXTRA CURRICULAR

1st Position at ANRF AISEHack 2026, Theme 2 Pollution Forecasting

Apr 2026

2nd Position at Yugantar 2025 (SIH Internal Round)

Oct 2025

Member & Volunteer, ACM Student Chapter — NMIMS MPSTME Shirpur

Oct 2025 - Present

- Volunteered at Initial Commit: AI Edition, an AI/ML workshop, assisting attendees with technical doubts.